Feature Completion Checklist

Project Title: AI Programming Tutor & Bug Explainer

1. Completed Features

The features below are fully implemented and working end to end across both the frontend and the backend. Each item is explained in detail so its scope and behaviour are clear.

User Authentication & Sessions — A complete account system with registration, login, and logout. Passwords are hashed with bcrypt before they ever reach the database, and login is verified against that hash. Sessions use signed JSON Web Tokens that are attached to every request and verified on the server, so a logged-in user stays authenticated across pages. Logout clears the stored token and returns the user to a guest state.

Role-Based Access (Student / Admin) — The platform supports two roles. Ordinary learners get the student role, while a configured admin-email list automatically grants admin rights on registration or login. Admin-only areas are protected on both the frontend (route guards) and the backend (middleware), so a student cannot reach admin functions even by typing the URL directly.

AI Coding Tutor — A question box where a learner can ask about any programming concept — arrays, recursion, OOP, binary search, and so on — and receive a simple, beginner-friendly answer with a definition, an everyday analogy, a small code example, and an encouraging note. It supports a mix of Bangla and English so beginners are never lost.

Bug Explainer — The learner pastes code into a Monaco (VS Code-style) editor, selects the language (C, C++, Java, Python, or JavaScript), and the AI explains what is wrong, points to the exact lines, gives the corrected code, and adds tips to avoid the same mistake next time.

Compiler & Runtime Error Translator (Flagship) — This tool turns scary error messages such as “Segmentation fault”, “NullPointerException”, or “Time Limit Exceeded” into plain, easy language — what the error really means, why it usually happens, and how to fix it. Common errors are available as one-click examples.

Code Improver — The learner pastes working code and receives friendly suggestions for cleaner variable names, simpler structure, lower time complexity, and better habits, along with an optional improved version — teaching good practice rather than just rewriting the code.

GitHub Repository Explainer — The learner pastes any public GitHub link, and the app reads the real repository through the GitHub API — its description, README, and a cleaned-up file list — then explains what the whole project does and how it is organized. Clicking any file gives a beginner-friendly, step-by-step explanation of that file.

PDF & Image Explainer — Learners can upload lecture slides, notes, textbook pages, or photos of handwritten code and problems by dragging a file or clicking to browse. Text-based PDFs are read directly, while images are understood by a vision model, and the AI explains the content simply. Uploads are limited to safe file types and sizes with clear messages when something is wrong.

Voice Tutor — Every AI explanation can be read aloud using the browser's built-in speech engine. The reader cleans away formatting symbols so the voice sounds natural, and a single button toggles between listening and stopping. It is free for all users, including guests.

Memory-Based Personalization — After each interaction the tutor stores a short note about what the learner worked on, and a compact summary of recent activity is quietly fed into later prompts. This lets the answers become more tailored to the individual learner over time, which is the main difference from an ordinary chatbot.

XP, Levels & Progress — Learners earn experience points for using the tools, and their level rises automatically as points accumulate. A profile page shows their name, role, plan, total XP, current level, and a progress bar toward the next level.

Analytics Dashboard — A personal dashboard shows XP, level, streak, and total activity at a glance, an "activity by tool" bar chart, quick links to jump back into any tool, and a list of recent history — giving each learner a clear snapshot of their learning.

Admin Panel — Administrators get a control panel with platform statistics (total users, number of admins, total AI activities, and top users by XP), a tool-usage chart, the full user list with the ability to promote or demote roles, and a live feed of recent activity across the whole platform.

AI Reliability & Security Layer — Behind the scenes, a single service connects the tools to the AI provider and hides the details. If a provider is slow, unavailable, or returns nothing, the request safely falls back to a built-in demo response instead of breaking, and a request timeout prevents the app from hanging. All AI keys are kept on the server and never exposed to the browser. The API is protected by rate limiting, cross-origin configuration, and a central error handler that returns clear, friendly messages.

2. In-Progress Features

Two features are partially implemented and are currently being completed. For each, the reason for incompletion and the planned completion date are given below.

Weak-Topic Tracking — In Progress. The data model, its unique index, and the logic that reads a learner's weak topics into the AI personalization context are all in place. What remains is the runtime write path that creates and increments a weak-topic record whenever a learner repeatedly struggles with a concept, so the collection is not yet populated automatically during normal use. This connecting logic is being added next.

Practice Problem Generator — In Progress. The backend endpoint and its prompt are complete and already return a structured practice problem, but the user-facing Practice Arena page that calls this endpoint is still being built, so the feature is not yet reachable from the interface. The page and its wiring to the existing endpoint are in progress.

3. Not-Started Features

Status: None. All planned functions of the project have already been started and are present in the codebase in some form, so there are no features remaining in the “Not Started” category at this stage. The only outstanding work is completing the two in-progress features listed above.